Habitat Detective Dataset

Story: Scientists checked places for **Azure Warblers** (fictional). **Yes** = nesting pair found.

INSTRUCTIONS

1. Study the training sites (A–J). Look for patterns.
2. Write a short rule: when do you predict a nest?
3. Predict mystery sites X, Y, and Z *before* answers are revealed.
4. Discuss what went wrong if your rule fails.

Training sites

Site	Temp	Tree cover	Insects	Human disturbance	Nest?
A	Cool	High	High	Low	Yes
B	Hot	Low	Low	High	No
C	Mild	High	High	Low	Yes
D	Warm	Medium	Medium	High	No
E	Cool	High	High	Low	Yes
F	Mild	High	Medium	Low	Yes
G	Hot	Medium	Low	Medium	No
H	Mild	Low	High	Low	No
I	Cool	Medium	High	Medium	No
J	Mild	High	High	Medium	Yes

Our rule

Mystery sites — predict first

Site	Temp	Tree cover	Insects	Human disturbance	Our prediction
X	Cool	High	High	Low	
Y	Hot	High	High	Low	
Z	Mild	High	High	High	

Student side — mystery answers on page 2 (teacher)

TEACHER KEY — REVEAL AFTER VOTES

Mystery Site Answers

Site	Nest?	Teaching note
X	Yes	Fits the cool/high cover/high insects/low disturbance pattern.
Y	No	Too hot — temperature matters even with good cover.
Z	Yes	Exception: high disturbance but nest boxes in a busy park (hidden factor).

Facilitation tips: Students often ignore site H (insects high but cover low) and site J (medium disturbance still OK). Use Z to launch “model ≠ reality.”